

Please make sure that you show all of the steps, not just the answer.

1) Use the derivative to find all intervals where $y = \sin^2(x) - \cos(x)$ is increasing on the interval $[0, 2\pi]$.

$$y' = 2 \sin x \cdot \cos x + \sin x$$

$$y' = \underbrace{\sin x}_{\downarrow} \underbrace{(2 \cos x + 1)}_{\downarrow} > 0$$

$$\sin x = 0 \quad 2 \cos x + 1 = 0$$

On $[0, 2\pi]$ the critical numbers are:

$$x = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, 2\pi$$

test intervals:

$$\left(0, \frac{2\pi}{3}\right), \left(\frac{2\pi}{3}, \pi\right), \left(\pi, \frac{4\pi}{3}\right), \left(\frac{4\pi}{3}, 2\pi\right)$$

pos neg pos neg

The function is increasing on
 $\left(0, \frac{2\pi}{3}\right) \cup \left(\pi, \frac{4\pi}{3}\right)$

2) Use the second derivative to find all intervals where $y = xe^{-x^2}$ is concave up.

$$y' = e^{x^2} + 2x^2 e^{x^2}$$

$$= e^{x^2} (1 + 2x^2)$$

$$y'' = e^{x^2} (2x) (1 + 2x^2) + e^{x^2} (4x)$$

$$= e^{x^2} [2x(1 + 2x^2) + 4x]$$

$$= e^{x^2} (4x^3 + 6x)$$

$$= 2x e^{x^2} (2x + 3)$$

$$\text{Since } e^{x^2} > 0 \text{ and } 2x^2 + 3 > 0$$

for all x , then the sign of y'' depends
 on x . So $y'' > 0$ when $x > 0$

Concave up on $(0, \infty)$

3) Use derivatives to find the intervals where $f(x) = x^3 - 7x^2 + x + 4$ is increasing and decreasing.

$$f'(x) = 3x^2 - 14x + 1 = 0$$

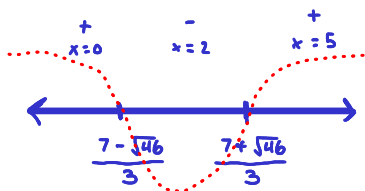
$$x = \frac{14x \pm \sqrt{196 - 12}}{6} = \frac{7 \pm \sqrt{46}}{3}$$

$$f'(0) = 1 > 0$$

$$f'(2) = -15 < 0$$

$$f'(5) = 6 > 0$$

$f(x)$ increasing on $\left(-\infty, \frac{7 - \sqrt{46}}{3}\right) \cup \left(\frac{7 + \sqrt{46}}{3}, \infty\right)$
 and decreasing on $\left(\frac{7 - \sqrt{46}}{3}, \frac{7 + \sqrt{46}}{3}\right)$



4) A rectangular box with a square base must have a volume of 12 cubic feet. Find the dimensions needed to construct the box with the least amount of material. The box does have a top and a bottom as well as 4 sides.

x = Side length of square base, h = height, S = surface area

$$x^2 h = 12$$

$$h = \frac{12}{x^2}$$

$$\begin{aligned} S &= 2x^2 + 4xh \\ &= 2x^2 + 4x \left(\frac{12}{x^2} \right) \\ &= 2x^2 + \frac{48}{x} \end{aligned}$$

$$S' = 4x - \frac{48}{x^2}$$

$$S' = 4x - \frac{48}{x^2} = 0$$

$$4x = \frac{48}{x^2}$$

$$4x^3 = 48$$

$$x^3 = 12$$

$$x = \sqrt[3]{12}$$

$$h = \frac{12}{x^2} = \frac{12}{(\sqrt[3]{12})^2} = \sqrt[3]{12}$$

Dimensions:

$$\sqrt[3]{12} \text{ ft} \times \sqrt[3]{12} \text{ ft} \times \sqrt[3]{12} \text{ ft}$$

5) Find the point on the parabola $y = x^2$ that is closest to the point (4,0).
Hint: you will need to use the formula for the distance between two points.

Point on parabola: (x, x^2)

Distance formula from (x, x^2) to $(4, 0)$

$$d = \sqrt{(x-4)^2 + (x^2-0)^2}$$

$$d = \sqrt{(x-4)^2 + x^4}$$

$$d^2 = \left(\sqrt{(x-4)^2 + x^4} \right)^2$$

$$D = (x-4)^2 + x^4$$

$$D = x^2 - 8x + 16 + x^4$$

$$D = x^4 + x^2 - 8x + 16$$

$$D' = 4x^3 + 2x - 8 = 0$$

↓

$$2x^3 + x - 4 = 0 \quad \text{Now Solve for } x$$

The derivative does not factor nicely

So, using a calculator:

$$x \approx 1.107$$

$$y = x^2 \approx (1.107)^2$$

$$\approx 1.226$$

Point: $(1.107, 1.226)$